LECTURE 17

Definition 0.1 *The expectation of a non negative random variable X is defined as*

$$EX \;=\; \sup\{EY \,|\, Y \text{ is simple and } Y \le X\}\,. \tag{0.1}$$

Remark 0.1 *From Lemma 1, Lecturenotes 16, we have*
(i)

$$EX \;=\; \lim_{n\to\infty} EX_n\,,$$

where X_n is a sequence of simple random variables as in Theorem 3, Lecturenotes 16.

(ii) If X_1, X_2 be non negative random variables such that $X_1 = X_2$ a.s., then $EX_1 = EX_2$.

Theorem 0.1 *Let X be a continuous non negative random variable with pdf f. Then*

$$EX \;=\; \int_0^\infty x f(x)\,dx$$

provided the rhs integral exists.

Proof. By using the simple random variables given in the proof of Theorem 3, Lecturenotes 16 we get

$$\begin{aligned}
EX &= \lim_{n\to\infty} EX_n \\
&= \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \frac{k}{2^n}\, P\Big(\frac{k}{2^n} \le X(\omega) < \frac{k+1}{2^n}\Big) \\
&= \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \Big[F((k+1)/2^n) \;-\; F(k/2^n)\Big] \\
&= \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} f(t)dt \\
&= \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} t f(t)dt \;+\; \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} (k/2^n - t) f(t)dt \\
&= \lim_{n\to\infty} \int_0^n t f(t)dt \;+\; \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} (k/2^n - t) f(t)dt.
\end{aligned} \tag{0.2}$$

Consider

$$\Big| \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} (k/2^n - t) f(t) dt \Big| \;\leq\; \frac{1}{2^n} \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} f(t) dt$$
$$= \; \frac{1}{2^n} \int_0^n f(t) dt \leq \frac{1}{2^n}.$$

Therefore

$$\lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \int_{\frac{k}{2^n}}^{\frac{k+1}{2^n}} (k/2^n - t) f(t) dt \;=\; 0.$$

Hence by substituting the above in (0.2), we get

$$EX \;=\; \int_0^\infty x f(x)\, dx\,.$$

Definition 0.2 *Let X be a random variable on $(\Omega, \mathcal{F}, P)$. The mean or expectation of X is said to exists if either EX^+ or EX^- is finite. In this case EX is defined as*

$$EX \;=\; EX^+ \,-\, EX^-\,,$$

where

$$X^+ \;=\; \max\{X, 0\},\; X^- \;=\; \max\{-X, 0\}\,.$$

Note that X^+ is the positive part and X^- is the negative part of X.

Theorem 0.2 *(Reading exercise) Let X be a continuous random variable with finite mean and pdf f. Then*

$$EX \;=\; \int_{-\infty}^\infty x f(x)\, dx\,.$$

Proof. Set

$$Y_n(\omega) \;=\; \begin{cases} \dfrac{k}{2^n} & \text{if} \quad \dfrac{k}{2^n} \leq X^+(\omega) < \dfrac{k+1}{2^n}\,, k = 0, \cdots, n2^n - 1 \\ 0 & \text{if} \quad X^+(\omega) \geq n\,. \end{cases}$$

Hence

$$Y_n(\omega) \;=\; \begin{cases} \dfrac{k}{2^n} & \text{if} \quad \dfrac{k}{2^n} \leq X(\omega) < \dfrac{k+1}{2^n}\,, k = 0, \cdots, n2^n - 1 \\ 0 & \text{if} \quad X(\omega) \geq n \text{ or } X(\omega) \leq 0\,. \end{cases}$$

Then Y_n is a sequence of simple random variables such that

$$EX^+ \;=\; \lim_{n\to\infty} EY_n\,.$$

Similarly, set

$$Z_n(\omega) \;=\; \begin{cases} \dfrac{k}{2^n} & \text{if} \quad \dfrac{k}{2^n} \le X^-(\omega) < \dfrac{k+1}{2^n}\;, k = 0, \cdots, n2^n - 1 \\ 0 & \text{if} \quad X^-(\omega) \ge n\,. \end{cases}$$

Hence

$$Z_n(\omega) \;=\; \begin{cases} \dfrac{k}{2^n} & \text{if} \quad \dfrac{k}{2^n} \le -X(\omega) < \dfrac{k+1}{2^n}\;, k = 0, \cdots, n2^n - 1 \\ 0 & \text{if} \quad X(\omega) \le -n \text{ or } X(\omega) \ge 0\,. \end{cases}$$

Then

$$EX^- \;=\; \lim_{n\to\infty} EZ_n\,.$$

Now

$$\lim_{n\to\infty} EY_n \;=\; \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \frac{k}{2^n} P(\{\frac{k}{2^n} \le X(\omega) < \frac{k+1}{2^n}\}) \tag{0.3}$$

and

$$\begin{aligned} -\lim_{n\to\infty} EZ_n \;&=\; -\lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \frac{k}{2^n} P\Big(\frac{k}{2^n} \le -X(\omega) < \frac{k+1}{2^n}\Big) \\ &=\; \lim_{n\to\infty} \sum_{k=0}^{n2^n-1} \Big(-\frac{k}{2^n}\Big) P\Big(\frac{-k-1}{2^n} < X(\omega) \le \frac{-k}{2^n}\Big) \\ &=\; \lim_{n\to\infty} \sum_{k=0}^{-n2^n+1} \frac{k}{2^n} P\Big(\frac{k-1}{2^n} < X(\omega) \le \frac{k}{2^n}\Big) \\ &=\; \sum_{k=0}^{-n2^n+1} \frac{k-1}{2^n} P\Big(\frac{k-1}{2^n} < X(\omega) \le \frac{k}{2^n}\Big)\,. \end{aligned} \tag{0.4}$$

The last equality follows by the arguments from the proof of 0.1. Combining (0.3) and (0.3), we get

$$EX \;=\; \lim_{n\to\infty} \sum_{k=-n2^n}^{n2^n-1} \frac{k}{2^n} P\Big(\frac{k}{2^n} \le X(\omega) < \frac{k+1}{2^n}\Big)\,.$$

Now as in the proof of Theorem 0.1, we complete the proof.

We state the following useful properties of expectation.

Theorem 0.3 *Let X, Y be random variables with finite mean. Then*

(i) If $X \geq 0$, then $EX \geq 0$.

(ii) For $a \in \mathbb{R}$,

$$E(aX + Y) \;=\; aEX \;+\; EY\,.$$

(iii) Let $Z \geq 0$ be a random variable such that $Z \leq X$. Then Z has finite mean and $EZ \;\leq\; EX$.

The proof of the above theorem is as follows.

(1) We have already seen the proofs of (i) and (ii) when X, Y are simple random variables. Proof of (iii) for simple random variables X and Z is an easy exercise.

(2) approximate non negative random variables using simple random variables and then use Lemma 1, Lecturenotes 16.

(3) For general random variable one use the decomposition $X = X^+ - X^-, Y = Y^+ - Y^-, Z = Z^+ - Z^-$.

Many results in probability theory uses the above technique, i.e. to prove a statement (involving expectation), first prove it for a class of random variables (which generally comes out to be easy) and then use convergence theorem involving expection, i.e.,

$$\lim_{n\to\infty} EX_n \;=\; E[\lim_n X_n].$$

For example, we already know that the above is true when $X \geq 0$, X_n's are simple and non negative and $X_n(\omega) \to X(\omega)$ for all $\omega \in \Omega$. In the context of Riemann integration, one can recall the following convergence theorem.

" If $g_n, n \geq 1$ is a sequence of continuous functions defined on the $[a, b]$ such that $g_n \to g$ uniformly in $[a,\ b]$, then

$$\lim_{n\to\infty} \int_a^b g_n(x)\,dx \;=\; \int_a^b g(x)dx\,."$$

i.e., to take limit inside the integral, one need uniform convergence of functions. In many situations, it is highly unlikely to get uniform convergence.

In fact uniform convergence is not required to take limit inside an integral. This is illustrated in the following theorems. The proof of the following theorem is an extension of the proof of Lemma 1, Lecturenotes 16. We leave it as an exercise for interested students.

Theorem 0.4 *(Monotone convergence theorem) Let X_n be a nondecreasing sequence of nonnegative random variables such that $\lim_{n\to\infty} X_n = X$. Then*

$$\lim_{n\to\infty} EX_n = EX\,.$$

[Here $\lim_{n\to\infty} X_n = X$ means $\lim_{n\to\infty} X_n(\omega) = X(\omega)$, $\omega \in \Omega$.]

Theorem 0.5 *(Fatou's lemma) Let $\{X_n\}$ be a sequence of non negative random variables. Then*

$$E\Big[\liminf_{n\to\infty} X_n\Big] \le \liminf_{n\to\infty} EX_n.$$

Proof: Recall that

$$\liminf_n X_n = \lim_{n\to\infty} Y_n,\ Y_n = \inf\{X_n, X_{n+1}, \cdots\}.$$

Then $Y_n \nearrow X := \liminf_n X_n,\ Y_n \le X_n$. Hence using monotone convergence theorem, we have

$$\begin{aligned} E\Big[\liminf_n X_n\Big] &= \lim_{n\to\infty} EY_n \\ &= \liminf_{n\to\infty} EY_n \\ &\le \liminf_{n\to\infty} EX_n. \end{aligned}$$

□

Theorem 0.6 *(Dominated Convergence Theorem) Let X_n, X, Y be random variables such that*

(i) Y has finite mean.

(ii) $|X_n| \le Y$.

(iii) $\lim_{n\to\infty} X_n = X$.

Then

$$\lim_{n\to\infty} EX_n \;=\; EX\,.$$

Proof: Given

$$-Y \le X_n \le Y, \text{ i.e. } Y - X_n \ge 0,\ X_n + Y \ge 0.$$

Hence using Fatou's lemma twice (by noting that $\liminf_{n\to\infty} X_n = \lim_n X_n$)we have

$$E[Y - X] \;=\; \liminf_{n\to\infty} E[Y - X_n] \le \;\; EY - \limsup_{n\to\infty} EX_n$$

and

$$E[Y + X] \le EY + \liminf_{n\to\infty} EX_n.$$

The above implies that

$$\limsup_{n\to\infty} EX_n \le EX \le \liminf_{n\to\infty} EX_n.$$

This completes the proof (why?) □

Example 0.1 *Let U be uniform $(0,\ 1)$ random variable (defined on $(\Omega, \mathcal{F}, P)$). Define*

$$X_n \;=\; nI_{\left\{U\in\left(0,\,\frac{1}{n}\right)\right\}},\ n \ge 1.$$

Observe that

$$X_n(\omega) \;=\; \begin{cases} n & \text{if} \qquad\quad \omega \in \left\{0 < U < \frac{1}{n}\right\} \\ 0 & \text{otherwise} \end{cases}$$

Then clearly $X_n(\omega) \to 0$ as $n \to \infty$ for all $\omega \in \Omega$ and $EX_n = 1$. Hence

$$\lim_{n\to\infty} EX_n \ne EX = 0, \text{ where } X \equiv 0.$$

Here $X \equiv 0$ means $X(\omega) = 0$ for all ω. Also note that there exists no random variable Y with finite mean such that

$$|X_n| \le Y.$$

Now we state the following theorem which provides a useful tool to compute expection of random variable which are mixed in nature.

Theorem 0.7 *Let X be a continuous random variable with pdf f and $\phi : \mathbb{R} \to \mathbb{R}$ be a continuous function.*

(i) If the integral $\int_{-\infty}^{\infty} \phi(x)f(x)dx$ is finite, then

$$E[\phi \circ X] \;=\; \int_{-\infty}^{\infty} \phi(x)f(x)dx\,.$$

(ii) If $\varphi \geq 0$, then

$$E[\phi \circ X] \;=\; \int_{-\infty}^{\infty} \phi(x)f(x)dx\,,$$

rhs integral may diverge to ∞.

Proof:(outline: proof of (i)) First, I will give a proof for a special case, i.e. when $\varphi : \mathbb{R} \to \mathbb{R}$ is strictly monotone and differentiable. Set $Y = \varphi(X)$. Then Y has density g given by

$$g(y) = \; f(\varphi^{-1}(y))|(\varphi^{-1})'(y)|, y \in \varphi(\mathbb{R}),\; = 0 \;\text{ otherwise.}$$

i.e.

$$g(y) = f(\varphi^{-1}(y))\Big|\frac{1}{\varphi'(\varphi^{-1}(y))}\Big|, y \in \varphi(\mathbb{R}),\; = 0 \;\text{ otherwise.}$$

Hence

$$\begin{aligned} E[\varphi(X)] &= \int_{-\infty}^{\infty} yg(y)dy \\ &= \int_{\varphi(\mathbb{R})} yf(\varphi^{-1}(y))\Big|\frac{1}{\varphi'(\varphi^{-1}(y))}\Big|dy \\ \text{(use } y = \varphi(x),\ \text{Jacobian is } \frac{dy}{dx} = \varphi'(x)) &= \int_{-\infty}^{\infty} \varphi(x)f(x)\Big|\frac{1}{\varphi'(x)}\varphi'(x)\Big|dx \\ &= \int_{-\infty}^{\infty} \varphi(x)f(x)dx. \end{aligned}$$

In particular, when $\varphi(x) = x^{2n+1}, n \geq 0$ and $E[X^{2n+1}]$ exists. Then note that φ is (strictly) increasing and differentiable. Hence

$$E[X^{2n+1}] \;=\; \int_{-\infty}^{\infty} x^{2n+1}f(x)dx.$$

When $\varphi(x) = x^{2n}$. Then note that φ is not one to one.

Set $Y = X^{2n}$ and G, g denote respectively the distribution function and the pdf of Y. Then for $y > 0$,

$$\begin{aligned} G(y) &= P\big(Y \le y\big) \\ &= P\big(-y^{\frac{1}{2n}} \le X \le y^{\frac{1}{2n}}\big) \\ &= F(y^{\frac{1}{2n}}) - F(-y^{\frac{1}{2n}}). \end{aligned}$$

Hence

$$g(y) \;=\; \frac{1}{2n} y^{\frac{1}{2n}-1}\Big(f(y^{\frac{1}{2n}}) + f(-y^{\frac{1}{2n}})\Big), y > 0, \;=\; 0 \text{ for } y \le 0.$$

Therefore

$$\begin{aligned} E[X^{2n}] &= \int_{-\infty}^{\infty} y g(y) dy \\ &= \frac{1}{2n}\int_0^{\infty} y^{\frac{1}{2n}}\Big(f(y^{\frac{1}{2n}}) + f(-y^{\frac{1}{2n}})\Big) dy \\ &= \frac{1}{2n}\int_0^{\infty} y^{\frac{1}{2n}} f(y^{\frac{1}{2n}}) dy + \frac{1}{2n}\int_0^{\infty} y^{\frac{1}{2n}} f(-y^{\frac{1}{2n}}) dy \\ (\text{use } y = x^{2n}) &= \int_0^{\infty} x f(x) x^{2n-1} dx + \int_0^{\infty} x f(-x) x^{2n-1} dx \\ (\text{use } u = -x) &= \int_0^{\infty} x^{2n} f(x) dx + \int_{-\infty}^{0} u^{2n} f(u) du \\ &= \int_{-\infty}^{\infty} x^{2n} f(x) dx. \end{aligned}$$

When $\varphi(x) = a_0 + a_1 x + \cdots + a_n x^n$, a polynomial of degree n, then using Theorem 0.3, we get

$$\begin{aligned} E\varphi(X) &= \sum_{k=0}^{n} a_k E X^k \\ &= \sum_{k=0}^{n} a_k \int_{-\infty}^{\infty} x^k f(x) dx \\ &= \int_{-\infty}^{\infty} \varphi(x) f(x) dx. \end{aligned}$$

Now suppose X is taking values in $[a,\ b]$ for some $-\infty < a < b < \infty$, the pdf f is continuous and $\varphi : [a,\ b] \to \mathbb{R}$ is continuous.

In this case, we use Weierstrass approximation theorem (i.e. there exists a sequence of polynomials p_n such that $p_n(x) \to \varphi(x)$ uniformly in $x \in [a,\ b]$.

Then $p_n(X_n(\omega) \to \varphi(X(\omega))\forall\omega$ (in fact the convergence is uniform in ω. Also

$$|p_n(X)| \leq K,$$

where $K = \sup_n \max\{p_n(x) : x \in [a,\ b]\}$. Hence using dominated convergence theorem,

$$\lim_{n\to\infty} Ep_n(X) = E\varphi(X).$$

Also using uniform convergence, it follows that

$$\lim_{n\to\infty} \int_a^b p_n(x)f(x)dx = \int_a^b \varphi(x)f(x)dx \ = \ \int_{-\infty}^{\infty} \varphi(x)f(x)dx.$$

Now from

$$Ep_n(X) = \int_a^b p_n(x)f(x)dx$$

and the above limits, we get

$$E\varphi(X) = \int_{-\infty}^{\infty} \varphi(x)f(x)dx$$

Now slightly more refined use of dominated convergence theorem, we can see the above result for any continuous φ and any X with a pdf.
Proof of (ii) follows by the above argument with monotone convergence theorem in place of DCT
□